

Schema Design Overview

Entity:

In order to allow tracing back to the original version of the collected data, we designed the table **Raw_Reviews**, which stores all attributes directly extracted from the App Store RSS feed, as well as several data quality monitoring attributes. These include 'duplicate_flag', which identifies duplicate records, and 'empty_after_cleaning', which indicates whether a review becomes empty after the text cleaning process. The auxiliary preprocessing attribute 'language' is used to identify how many non-English records should be removed. During the preprocessing process, we dropped many records to maintain a good-quality dataset; therefore, we also added an attribute: 'record_delete_reason' for tracking what records were deleted and why they were deleted. Although these attributes are not directly extracted from the RSS feed, they are intentionally stored in the **Raw_Reviews** table as preprocessing metadata rather than analytical data. This design allows every preprocessing decision to be traced, audited, and reproduced if necessary.

The **Cleaned_Reviews** table stores the 'cleaned content' attribute, which stores the non-empty review text after preprocessing and serves as the primary input for future sentiment analysis. The attribute 'date_transformed' stores the datetime format version of the original 'date' field, enabling time-based filtering and analysis. To reduce data redundancy, attributes that remain unchanged are referenced through the foreign key 'raw_review_pk' to the **Raw_Reviews** table instead of being duplicated in the **Cleaned_Reviews** table. We separate the cleaned data from the original records to preserve the integrity of the raw dataset while providing an analysis-ready dataset for downstream NLP tasks.

While reviewing the collected attributes, we found that **App** and **Country** each have a one-to-many relationship with review records. In addition, both entities contain additional metadata, and our initial sampling strategy also defined attributes such as 'app_id', 'app_category', and 'region', which were used during sampling but were not directly included in the collected RSS data. Therefore, we separated these attributes into the **App** and **Country** tables to reduce data redundancy, avoid update anomalies, and preserve metadata defined in our original sampling design.

Key Design:

For the **Raw_Reviews** table, 'review_id' is a natural key extracted from the App Store RSS feed; however, duplicate records may occur due to overlapping web scraping. Therefore, we introduced the surrogate key 'raw_review_pk' as the primary key to uniquely identify each record. Since each review belongs to one App and one Country, 'app_id' and 'country_id' are used as foreign keys. The **App** table uses 'app_id' as its natural primary key because it is the unique identifier assigned by the App Store, while the **Country** table uses the surrogate key **country_id** to provide a stable identifier and avoid update anomalies. The **Cleaned_Reviews** table uses 'clean_review_pk' as its surrogate primary key and 'raw_review_pk' as a foreign key to maintain traceability while avoiding unnecessary duplication of unchanged attributes.

Entity Relationship:

Unlike the one-to-many relationships between the **App** and **Country** tables and **Raw_Reviews**, each raw review corresponds to at most one cleaned review. Since some raw review records are removed during preprocessing to maintain a high-quality dataset, the relationship between **Raw_Reviews** and **Cleaned_Reviews** is 1:0 to 1.

Normalization:

The schema satisfies the principles of Third Normal Form (3NF) by ensuring that each table represents a single entity, every non-key attribute depends only on the primary key, and transitive dependencies are removed.